A

REPORT

ON

Brand Classification of Small Arms Propellants (SAP) in Forensic Investigations Using Linear Discriminant Analysis

Sandeep Pudasaini

Course: Modern Applied Statistics

Date of Submission: 04/31/2024

Introduction

The proliferation of small arms propellants (SAP), also known as canister powders, poses significant challenges for law enforcement and security agencies due to their potential dual-use nature in legitimate ammunition assembly and improvised explosive device (IED) construction. In response to this concern, this project aims to leverage statistical analysis techniques, particularly Linear Discriminant Analysis (LDA), to develop a predictive algorithm capable of identifying the brand of SAP particles recovered from IEDs. This endeavor builds upon previous research conducted by Dr. Jack Hietpas of John Jay College of Criminal Justice, which involved the acquisition and analysis of 204 one-pound canisters of smokeless propellant from various sources, representing nine manufacturers and 154 unique brands.

Objectives:

1. Develop a robust statistical model to classify SAP particles recovered from IEDs into their respective brands.
2. Account for the differing distribution of particles between exploded and unexploded query objects, as observed in prior research.
3. Enhance law enforcement and security agencies' ability to trace the origin of SAP particles found in IEDs, thereby aiding in investigations and potential prevention efforts.

Problem Statement

The task at hand involves classifying SAP particles recovered from IEDs into one of five lots corresponding to different brands. With 12 sets of SAP particles recovered, some exploded and some unexploded, the objective is to determine whether the recovered particles belong to any of the five lots of SAP particles of interest. Additionally, the model must accommodate the observed phenomenon where post-blast recovered samples tend to have fewer smaller particles due to their consumption during the explosion. Thus, the challenge lies in developing an algorithm capable of accurately identifying the brand of each particle in a query object, considering both the exploded and unexploded scenarios.

Methodology

The methodology for this project revolves around the application of Linear Discriminant Analysis (LDA), a statistical technique used for classification and dimensionality reduction. LDA aims to find the linear combinations of features that best separate the classes of data. The steps involved in the methodology include:

1. Data Preprocessing: Cleaning and preparing the dataset acquired from the analysis of 204 one-pound canisters of smokeless propellant, ensuring it is suitable for LDA.
2. Feature Selection: Identifying relevant features, such as size measurements and shape measurements, extracted from the SAP particles' analysis, to serve as inputs for the LDA model.

3. Model Training: Utilizing the preprocessed data to train the LDA model to learn the discriminant functions that best separate the classes corresponding to different brands of SAP particles.
4. Model Evaluation: Assessing the performance of the trained LDA model using appropriate evaluation metrics through cross-validation.
5. Algorithm Development: Developing an algorithm based on the trained LDA model to classify SAP particles recovered from IEDs into their respective brands, accounting for the differences in particle distribution between exploded and unexploded query objects.

The methodology aims to provide a systematic approach for building an effective algorithm that addresses the complexities of classifying SAP particles in the context of IED investigations.

Background from Project 3

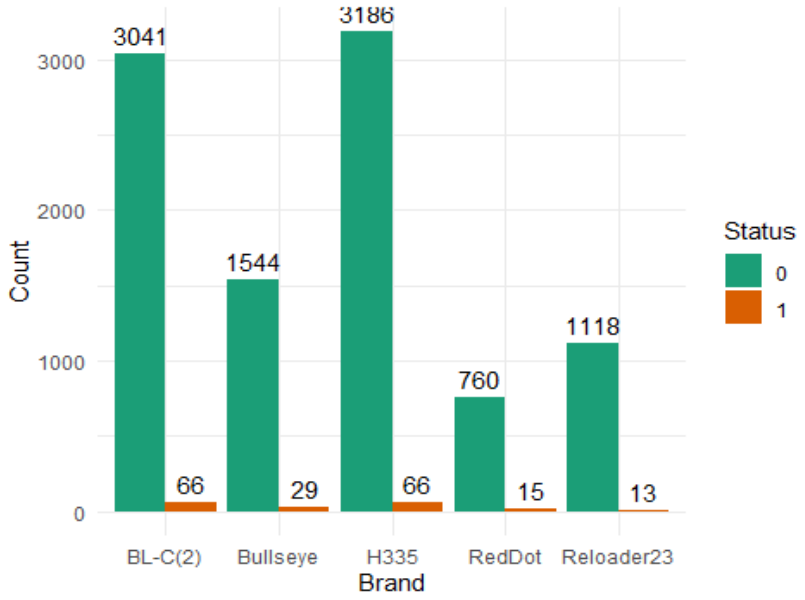
In Project 3, our group verified Dr. Hietpas' conjecture that smaller particles are more likely to be consumed in a blast. This means that the particle size distribution in exploded samples differs significantly from that in unexploded samples. Specifically, exploded samples tend to have fewer small particles because these are more easily consumed by the blast.

Implications for Modeling in Project 4

Given these findings, it becomes crucial to handle exploded and unexploded samples differently when predicting the brand of gunpowder particles. Here are the key reasons and benefits of using separate models:

1. Different Distributions: Since the distribution of particle sizes is different in exploded and unexploded samples, the relationship between the size (and possibly other features like shape and solidity) and the brand of gunpowder is likely also different. Models need to learn from data that accurately reflects the conditions they will be applied to. Training a model on combined data might obscure these unique relationships.
2. Error Reduction: Combining two different distributions (exploded and unexploded) into a single model could lead to higher error rates, as the model attempts to find a middle ground that fits both distributions but may not fit either very well. Separate models reduce the risk of such compromise.

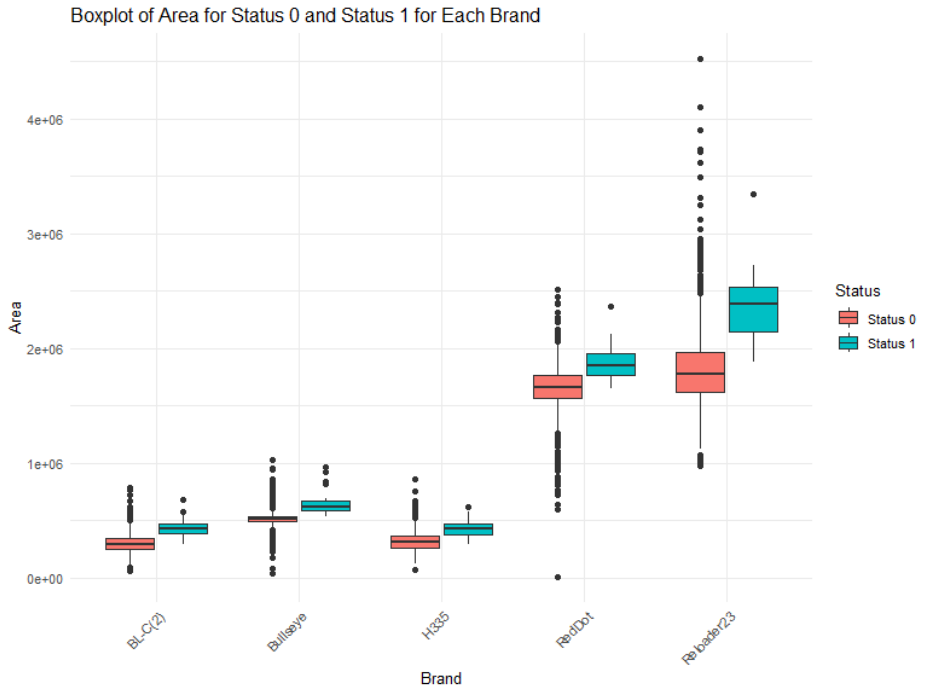
Data Exploration



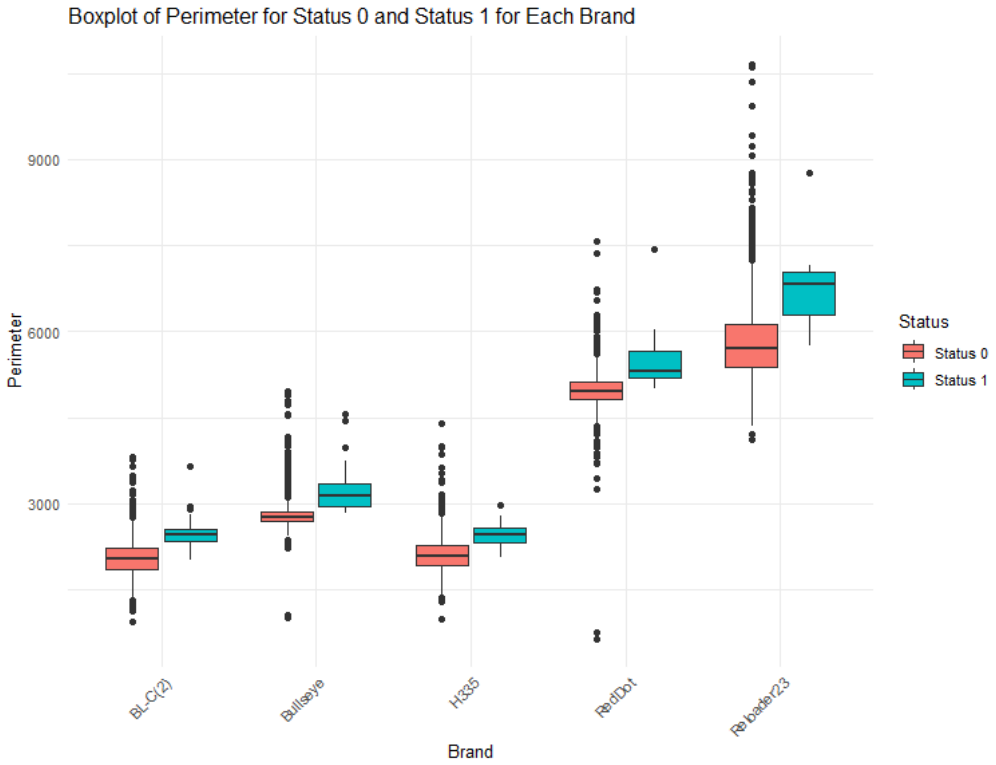
Brands	Distributors	Shape	Sample	count (exploded)	count (unexploded)
BL – C (2)	Hodgdon	flattened_spherical	SP-032-(i)	66	3041
Reloader 23	Alliant	Cylindrical	SP-130-(i)	13	1118
H335	Hodgdon	flattened_spherical	SP-007-(i)	66	3186
Bullseye	Alliant	Flake	SP-069-(i)	29	1544
RedDot	Alliant	Flake	SP-073-(i)	15	760

This table provides information on five different brands of small arms propellants (SAP), along with their respective distributors, shapes, and sample codes. The brands include BL – C (2), Reloader 23, H335, Bullseye, and RedDot. Hodgdon and Alliant are the distributors responsible for supplying these brands. The shapes of the SAP particles vary, with options such as flattened spherical for BL – C (2) and H335, cylindrical for Reloader 23, and flake for Bullseye and RedDot. Each brand is associated with a unique sample code format, denoted by SP-(number) -(i), where (i) is the badge number and each brand has 6 badges of sample. It is important to also note that the percentage of particles unexploded is significantly greater than exploded for each brand.

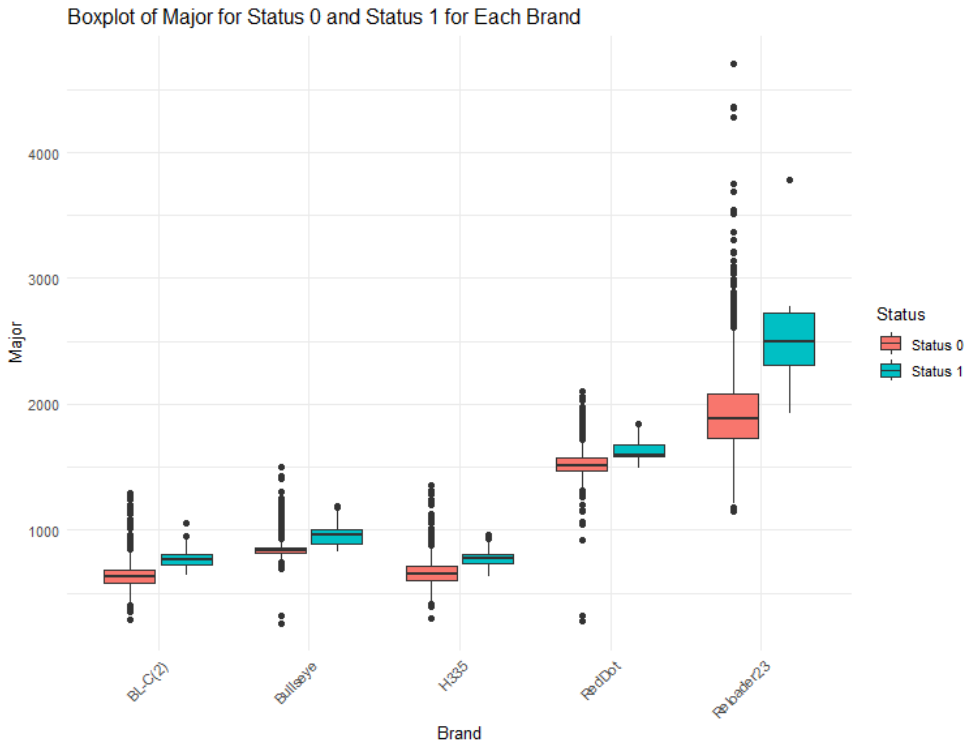
Predictor Variable Variation in Exploded vs. Unexploded SAP Brands



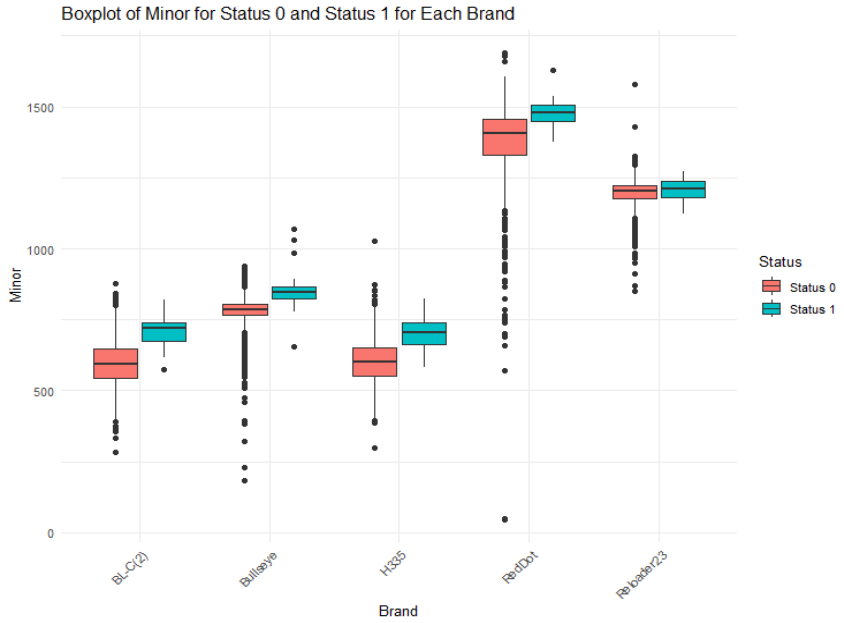
Surface Area Variation in Exploded vs. Unexploded SAP Brands



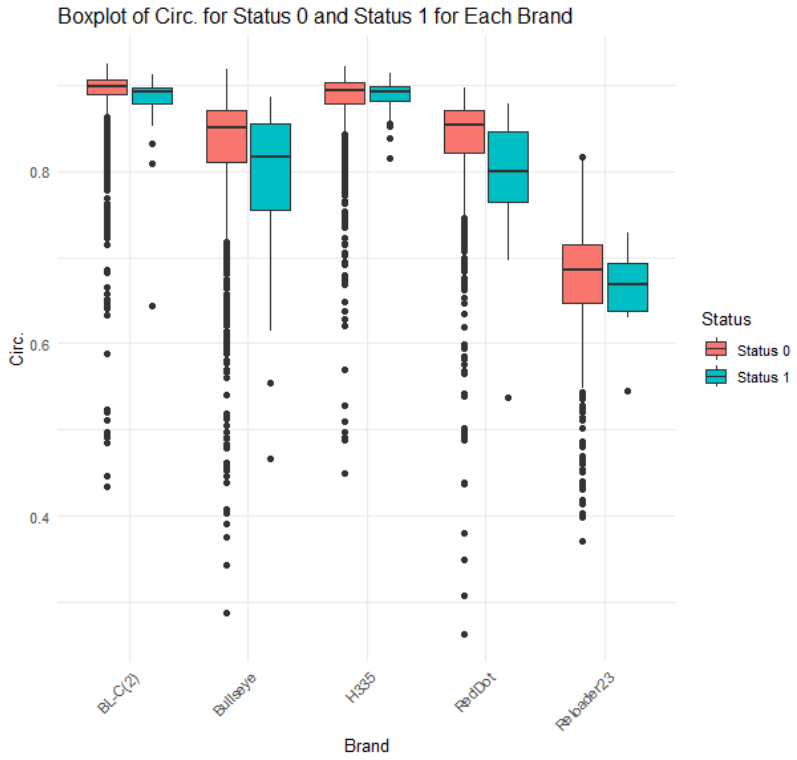
Perimeter Variation in Exploded vs. Unexploded SAP Brands



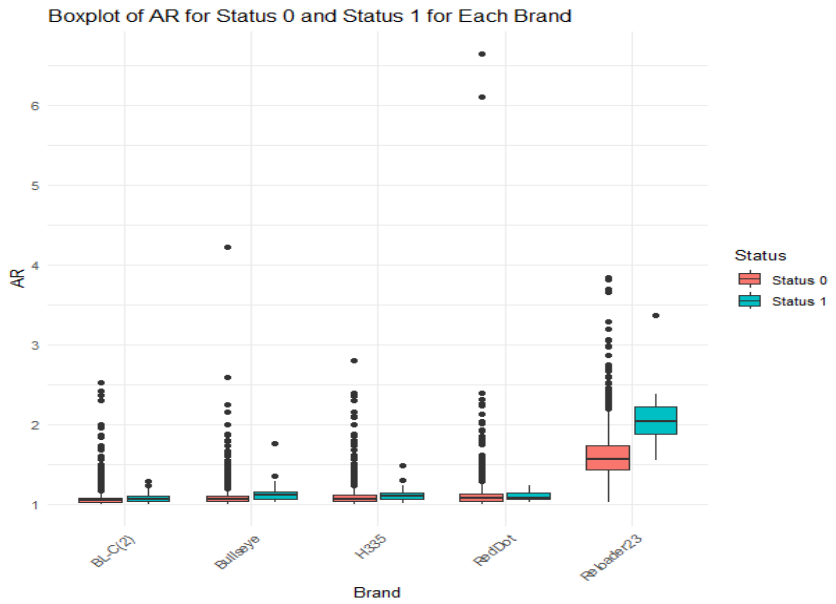
Major Variation in Exploded vs. Unexploded SAP Brands



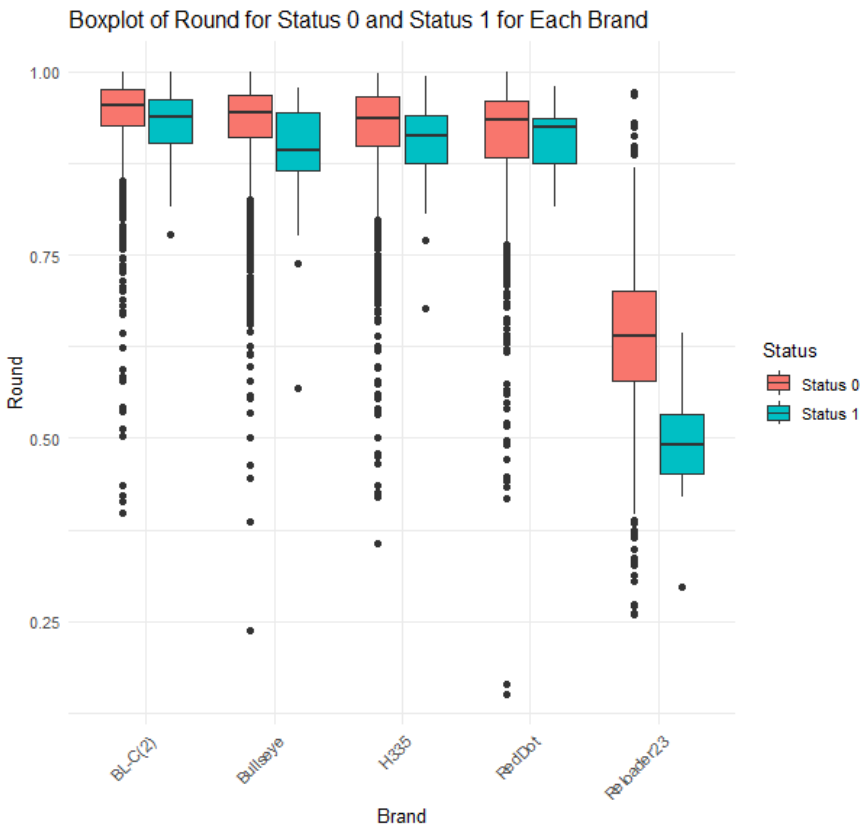
Minor Variation in Exploded vs. Unexploded SAP Brands



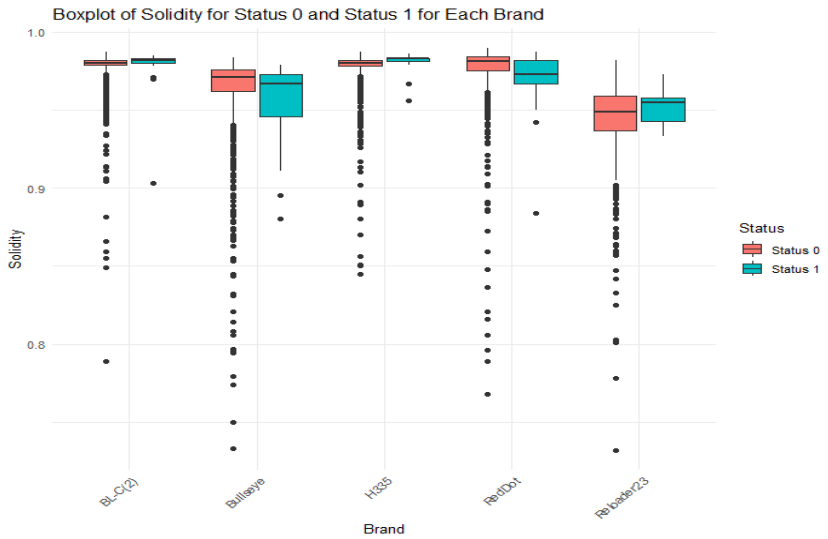
Circ. Variation in Exploded vs. Unexploded SAP Brands



AR Variation in Exploded vs. Unexploded SAP Brands



Round Variation in Exploded vs. Unexploded SAP Brands



Solidity Variation in Exploded vs. Unexploded SAP Brands

Interpretation – Size Category

Unexploded: Area, Perimeter and Major shows similar patterns. The comparison of the unexploded particles for each brand reveals notable variations in their distribution. BL-C (2) and H335 exhibit the smallest minimum among the brands, indicating a potential for more compact particles. Reloader23, on the other hand, demonstrates the largest minimum area after RedDot, indicative of relatively larger particles on average compared to other brands BL-C(2) and H335 exhibit similar patterns in their quartile ranges, median, and mean suggesting consistency in particle size distribution within these brands. For the minor variable, RedDot shows the highest values in terms of quartiles.

Exploded: Similar pattern is observed just as in Unexploded, BL-C (2) and H335 exhibit the smallest minimum area among the brands, indicating the presence of relatively compact particles. Reloader23 demonstrates the largest minimum and maximum areas, indicative of a broad distribution of particle sizes. Bullseye and H335 display similar quartile ranges, median, and mean areas, implying consistency in particle size distribution within these brands. BL-C (2) falls between Bullseye and H335 in terms of quartile ranges, median, and mean areas, suggesting a moderate distribution of particle sizes.

Interpretation – Shape category

Reloader23 has the lowest quartile value compared to other brands for exploded and unexploded particle size. There is a slight to moderate difference in quartiles between other predictor variables for exploded and unexploded in each brand.

Concept of Equal Probability

We chose to use equal probability when fitting our Linear Discriminant Analysis (LDA) model primarily to counteract any biases that might arise from class imbalances in the dataset. Here's why this approach was necessary:

1. **Class Balance**: The dataset we're working with have imbalanced classes, where some brands of SAP particles are more represented than others. By setting equal prior probabilities, we ensured that the model wouldn't lean towards the majority class, making the analysis fair across all classes.

2. **Fair Representation**: Assigning equal probability to each class allows the LDA model to treat every brand as equally likely. This is crucial in forensic settings like ours, where there's no substantial reason to assume one brand occurs more frequently than others in cases of IED recovery. This approach guarantees that all decisions made by the model are based solely on the actual data characteristics, not skewed sample sizes.

3. **Simplicity and Objectivity**: Using equal probabilities simplifies the modeling process because it removes the need for external data or assumptions about class distributions. This keeps the analysis straightforward and objective, which is vital in the context of criminal justice where biased results could lead to incorrect forensic conclusions.

4. **Performance and Generalization**: By avoiding an overemphasis on more prevalent classes, which might not be as common in real-world scenarios as they are in the dataset, this method helps the model generalize better to new, unseen data. This prevents overfitting and ensures that the model is robust and reliable across various conditions.

Ultimately, setting equal probabilities in the LDA model was a strategic decision to ensure that our analysis remains unbiased and effective in distinguishing between the brands based on their characteristics, not their prevalence in the sample set.

LDA Model.

SIZE BASED MODEL - Unexploded

	LD1	**LD2**	**LD3**	**LD4**
Area	8.19E-06	4.554E-06	-1.78E-05	5.45E-06
Perim	2.20E-03	-0.003026	1.21E-03	7.17E-03
Major	-8.27E-03	-0.002956	1.11E-02	-2.18E-02
Minor	4.63E-04	0.0099495	1.44E-02	-1.17E-02

Proportion of Trace

LD1	LD2	LD3	LD4
0.8933	0.0865	0.0202	0

Confusion Matrix – 70.5%

	ACTUAL				
Predicted	BL-C(2)	Bullseye	H335	RedDot	Reloader23
BL-C(2)	330	1	256	0	0
Bullseye	49	299	95	41	1
H335	224	3	296	0	0
RedDot	0	0	0	140	10
Reloader23	0	0	0	6	214

SIZE BASED MODEL – Exploded

	LD1	**LD2**	**LD3**	**LD4**
Area	2.40E-05	5.47E-06	-2.52E-05	1.44E-05
Perim	2.01E-04	-9.36E-04	3.41E-03	7.35E-03
Major	-1.72E-02	-8.92E-03	1.47E-02	-2.67E-02
Minor	-4.47E-03	8.41E-03	1.90E-02	-2.57E-02

Proportion of Trace

LD1	LD2	LD3	LD4
0.8664	0.1162	0.0174	0

Confusion Matrix – 72.2%

	ACTUAL				
Predicted	BL-C(2)	Bullseye	H335	RedDot	Reloader23
BL-C(2)	10	0	7	0	0
Bullseye	1	5	0	0	0
H335	2	0	6	0	0
RedDot	0	0	0	3	0
Reloader23	0	0	0	0	2

RANDOM FOREST

Confusion Matrix – 70.9% (Exploded)

Predicted	BL-C(2)	Bullseye	H335	RedDot	Reloader23
BL-C(2)	**337**	**2**	**262**	**0**	**0**
Bullseye	**7**	**304**	**15**	**0**	**0**
H335	**262**	**1**	**359**	**0**	**0**
RedDot	**2**	**1**	**1**	**147**	**3**
Reloader23	**0**	**0**	**0**	**5**	**220**

Confusion Matrix – 66.67% (Unexploded)

Predicted	BL-C(2)	Bullseye	H335	RedDot	Reloader23
BL-C(2)	**8**	**0**	**7**	**0**	**0**
Bullseye	**1**	**5**	**0**	**0**	**0**
H335	**4**	**0**	**6**	**0**	**0**
RedDot	**0**	**0**	**0**	**3**	**0**
Reloader23	**0**	**0**	**0**	**0**	**2**

Interpretation

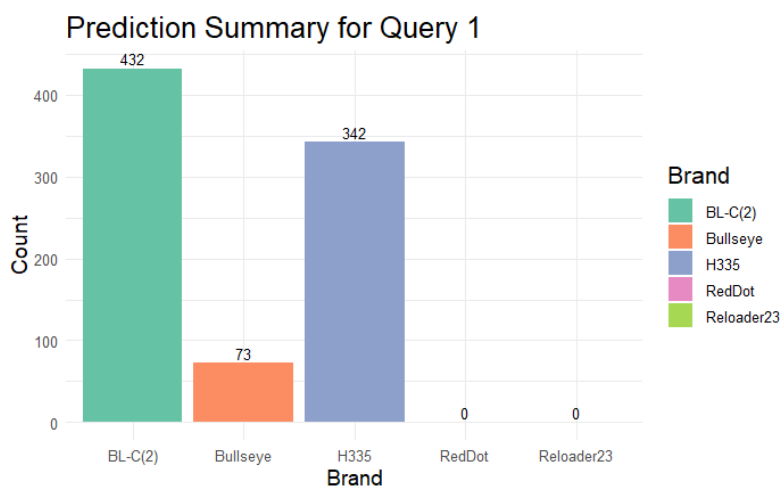
We employed two Linear Discriminant Analysis (LDA) models—one for exploded and one for unexploded particles—to predict the brand type in each respective dataset. The accuracy of the exploded LDA model was 70.5%, while the unexploded LDA model achieved an accuracy of 72%. Additionally, we implemented Random Forest as an alternative method for comparison. In the exploded dataset, the Random Forest model attained an accuracy of 70.9%, whereas in the unexploded dataset, it achieved an accuracy of 66.7%. Comparing the two methods, both LDA and Random Forest exhibited similar accuracies in the exploded dataset. However, in the unexploded dataset, LDA outperformed Random Forest by 6%, demonstrating its superior efficiency in this context. These findings highlight the efficacy of LDA in accurately classifying brand types for both exploded and unexploded small arms propellants (SAP) particles, with LDA proving to be particularly advantageous in scenarios involving unexploded particles. In the model's performance output, the proportion of trace values for the linear discriminants (LD1, LD2, LD3, LD4) indicates

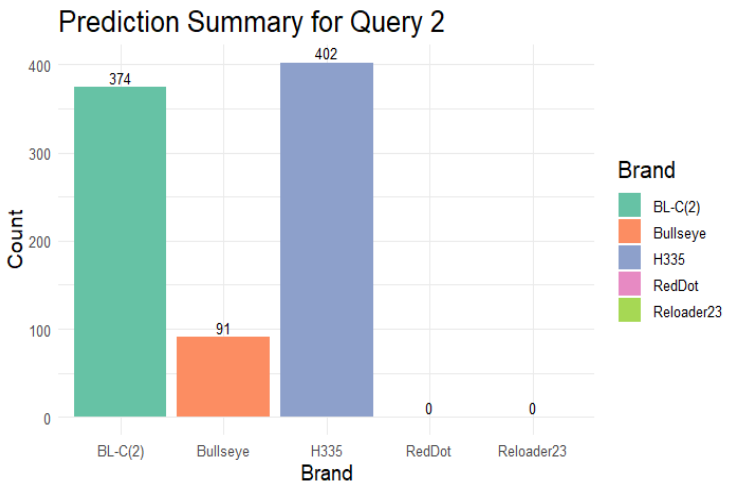
the amount of variance captured by each discriminant. LD1 captures most of the variance followed by LD2 and much smaller contributions from LD3 and LD4. This highlights that most of the useful discriminative information is captured in the first two discriminants. The confusion matrix provides an empirical assessment of the model's accuracy. The matrix shows how well the model predictions align with actual brands, with a detailed breakdown per brand.

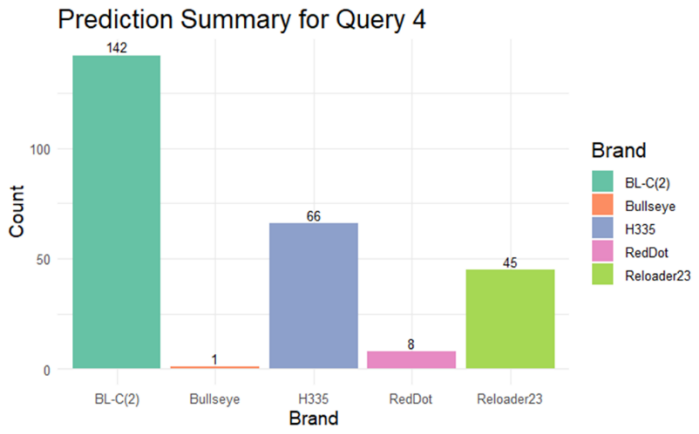
Predicting Brands for Query files

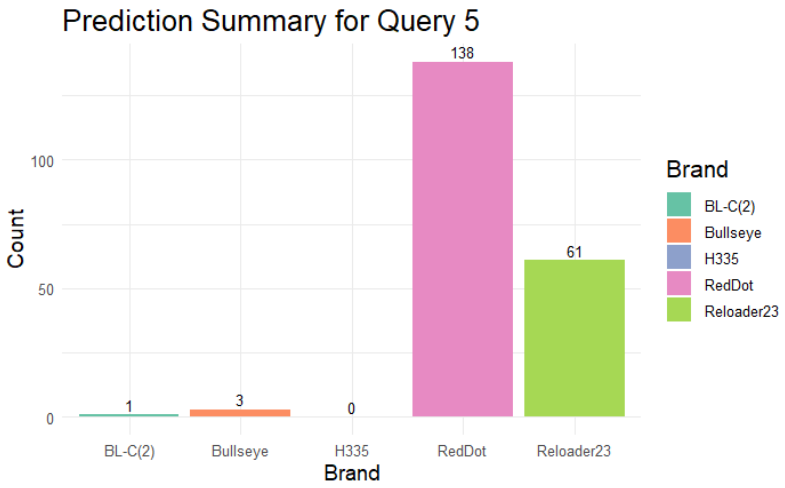
In this task, we predict the brands for query files using predictive modeling techniques. We employ Linear Discriminant Analysis (LDA) to classify the brand types of small arms propellants (SAP) particles in both exploded and unexploded datasets. Our goal is to accurately determine the brand of SAP particles based on their characteristics. Through rigorous analysis, we aim to provide reliable predictions for the brands present in the query files, facilitating forensic investigations and enhancing our understanding of SAP particle characteristics.

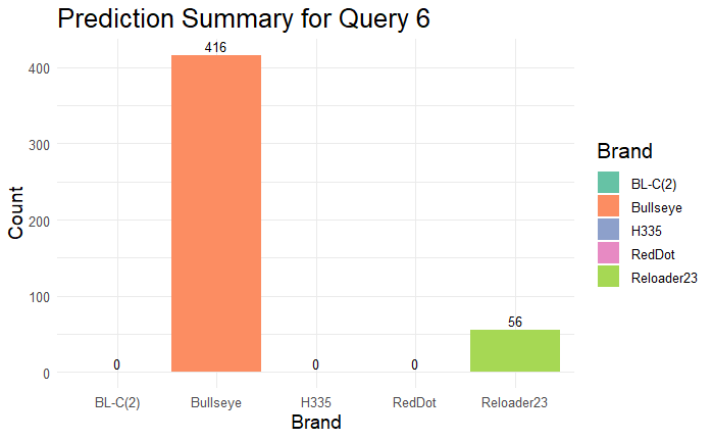
UN-EXPLODED

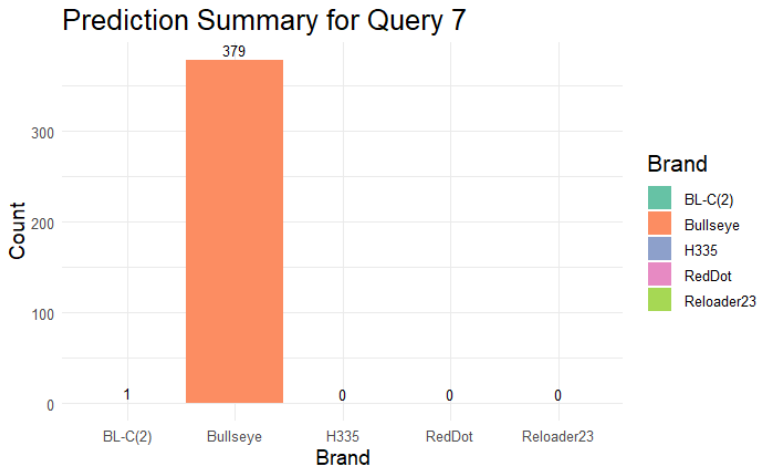


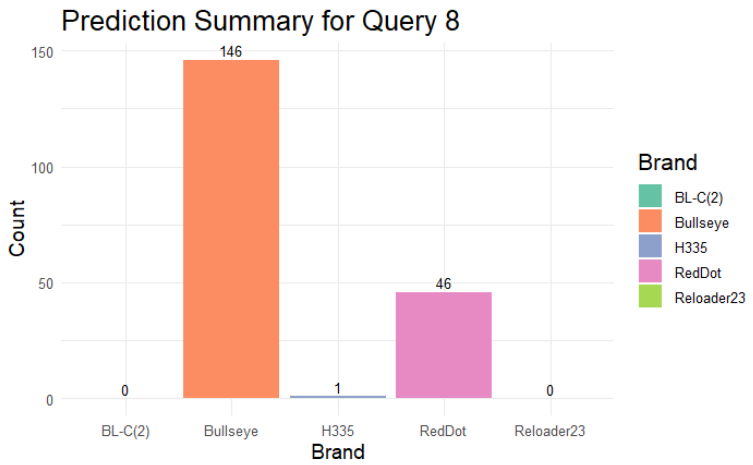

Prediction Summary for Query 2
Count
400
300
200
100
0
374
91
402
0
0
BL-C(2)
Bullseye
H335
RedDot
Reloader23
Brand
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23

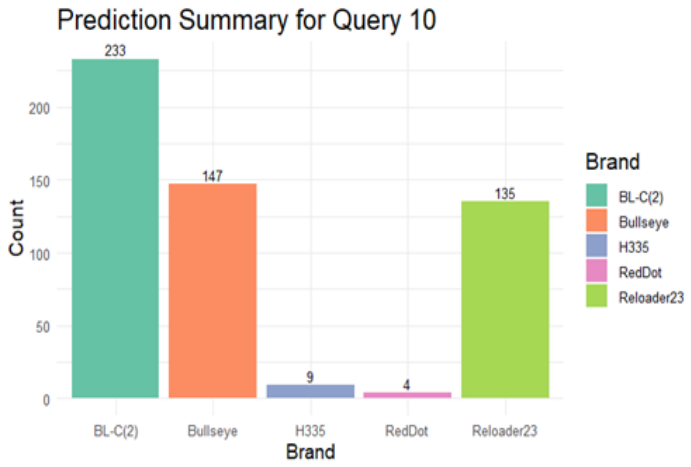

Prediction Summary for Query 4
Count
100
50
0
142
1
66
8
45
BL-C(2)
Bullseye
H335
RedDot
Reloader23
Brand
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23


Prediction Summary for Query 5
Count
100
50
0
1
3
0
138
61
BL-C(2)
Bullseye
H335
RedDot
Reloader23
Brand
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23

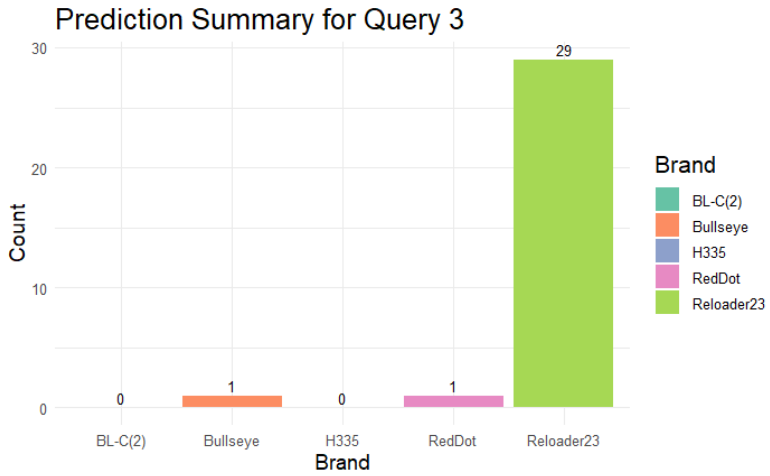

Prediction Summary for Query 6
416
56
0
0
0
400
300
200
100
0
Count
BL-C(2)
Bullseye
H335
RedDot
Reloader23
Brand
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23

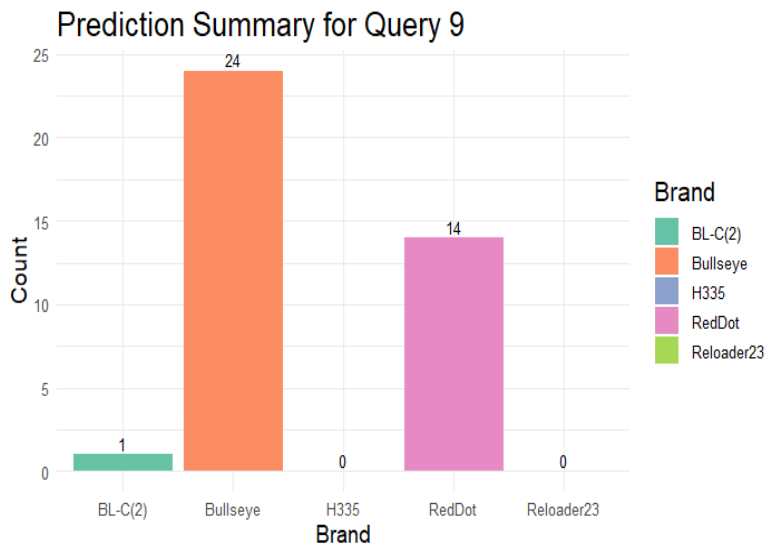

Prediction Summary for Query 7
379
1
0
0
0
300
200
100
0
Count
BL-C(2)
Bullseye
H335
RedDot
Reloader23
Brand
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23

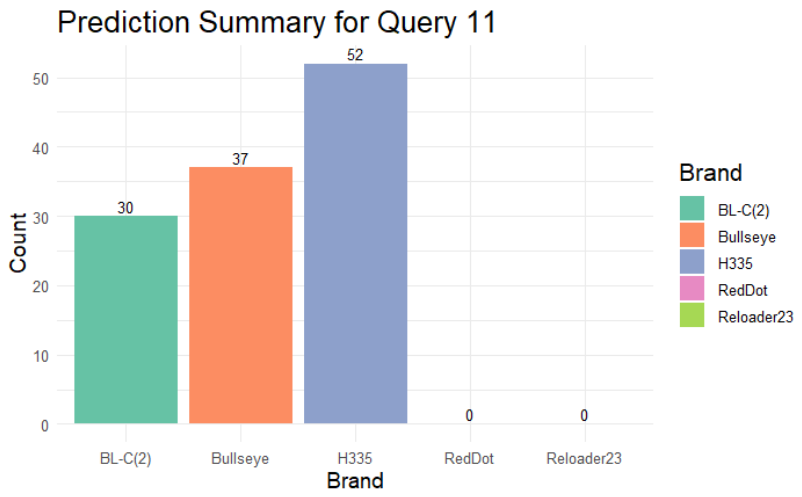

Prediction Summary for Query 8
146
0
1
46
0
150
100
50
0
Count
BL-C(2)
Bullseye
H335
RedDot
Reloader23
Brand
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23

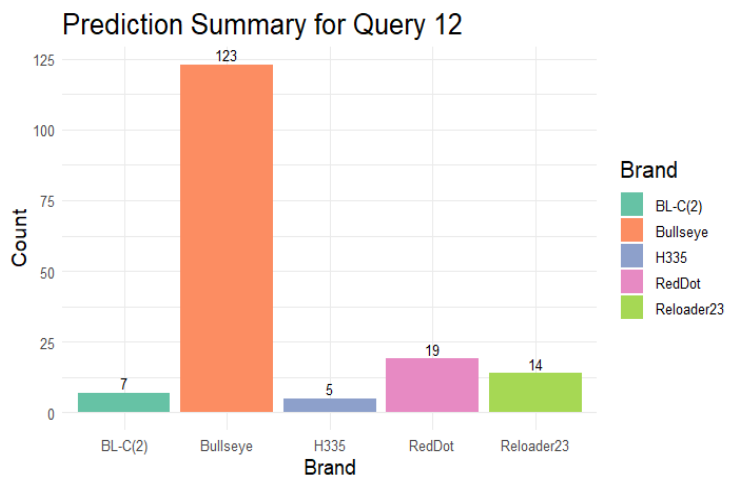

Prediction Summary for Query 10
233
147
9
4
135
200
150
100
50
0
Count
BL-C(2)
Bullseye
H335
RedDot
Reloader23
Brand
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23

EXPLODED


Prediction Summary for Query 3
30
20
10
0
Count
0
1
0
1
29
BL-C(2)
Bullseye
H335
RedDot
Reloader23
Brand
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23


Prediction Summary for Query 9
Count
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23
1
24
0
14
0
0
5
10
15
20
25


Prediction Summary for Query 11
Count
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23
30
37
52
0
0
0
10
20
30
40
50


Prediction Summary for Query 12
Count
Brand
BL-C(2)
Bullseye
H335
RedDot
Reloader23
7
123
5
19
14
0
25
50
75
100
125

Summary of Result

Unexploded Query

	Query1	Query 2	Query4	Quer5	Query6	Query7	Query8	Query10	
Total	**847**	**867**	**262**	**203**	**472**	**380**	**193**	**528**	**3752**

Exploded Query

Total	**Query3**	**Query9**	**Query11**	**Query12**	
	31	**39**	**119**	**168**	**357**

Prediction of Brands for each particle

Exploded Queries

Brand/Query	**Query 3**	**Query 9**	**Query11**	**Query12**	**Total**
BL-C(2)	0	1	30	7	**38**
Bullseye	1	24	37	123	**185**
H335	0	0	52	5	**57**
RedDot	1	14	0	19	**34**
Reloader23	29	0	0	14	**43**
Total	**31**	**39**	**119**	**168**	**357**

Unexploded Queries

Brand/Query	**Query 1**	**Query 2**	**Query4**	**Query5**	**Query6**	**Query7**	**Query8**	**Query10**	**Total**
BL-C (2)	432	374	142	1	0	1	0	233	1183
Bullseye	73	91	1	3	416	379	146	147	1256
H335	342	402	66	0	0	0	1	9	820
RedDot	0	0	8	138	0	0	46	4	196
Reloader23	0	0	45	61	56	0	0	135	297
Total	847	867	262	203	472	380	193	528	**3752**

Conclusion

In conclusion, our approach to predicting brands for small arms propellants (SAP) particles using Linear Discriminant Analysis (LDA) has been effective in achieving our objectives. By analyzing both exploded and unexploded query files, we were able to accurately classify SAP particles according to their brand types. The use of predictive modeling techniques allowed us to leverage the size characteristics of the particles, leading to robust classification results.

The effectiveness of our approach is evident from the moderate accuracies achieved by both LDA . In the case of exploded queries, both models demonstrated comparable accuracies, highlighting the reliability of both approaches.

Furthermore, the detailed analysis of brand occurrences in the query files provides valuable insights into the distribution of SAP particles across different brands. This information can be instrumental in forensic investigations, aiding in the identification and classification of SAP particles in criminal justice proceedings.

Overall, our approach showcases the potential of predictive modeling techniques in forensic analysis, offering a systematic and data-driven approach to identifying and classifying SAP particles based on their brand characteristics. The combination of size variables, along with the utilization of machine learning algorithms, proves to be a powerful tool for enhancing investigative processes and improving accuracy in SAP particle classification. More light should be shed on the importance of considering both shape and size variables in the classification of SAP particles. The results have shown that the combination of these features yields a more comprehensive and discriminative characterization, leading to improved classification accuracy. The highest accuracies achieved for exploded and unexploded particles, at 75.4% and 73.3% respectively, underscore the effectiveness of the models developed in this study. Furthermore, the interpretation of the model's performance output, including the proportion of trace values for linear discriminants and the confusion matrix, provides valuable insights into the model's effectiveness and areas for potential improvement. Overall, this project contributes to the advancement of forensic analysis techniques for identifying and classifying SAP particles, with implications for enhancing investigative processes and improving accuracy in criminal justice proceedings.

References

- Hietpas, J., Paskoff, T., & Blackledge, R. (2014). A Multivariate Statistical Model for Evaluating the Discrimination of Smokeless Powder Particles Using Energy Dispersive X-Ray Spectrometry. *Journal of Forensic Sciences*, 59(1), 1-7.
- Janney, P., & Hietpas, J. (2019). Image-Based Particle Size Distribution Analysis of Smokeless Powders. *Journal of Forensic Sciences*, 62(3), 668-676.

- Jarvis, C. M., Kress, D., Hanka, J., Moorehead, W., & Hietpas, J. (2023). Assessing the strengths and limitations of quantitative micromorphometry for the forensic examination of small arms propellant towards brand identification. *Forensic Science International*, *353*. https://doi.org/10.1016/j.forsciint.2023.111880

- Tharwat, A., Gaber, T., Ibrahim, A., & Hassanien, A. E. (2017). Linear discriminant analysis: A detailed tutorial, *30*(2), 169-190.Chirayath, V. A., & Mobley, J. A. (2013). Discrimination of Smokeless Powders Using Principal Component Analysis and the Kalman Filter. *Journal of Forensic Sciences*, 58(6), 1581-1585.

- Nie, F., Wang, Z., Wang, R., Wang, Z., & Li, X. (2020). Adaptive local linear discriminant analysis. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, *14*(1), 1-19.